Ratios and matching amounts

25 questions, including 2 challenges. Write your work and answers on separate paper.

1. 4.2×10

2. $400 \div 100$

3. Double 4.5

4. Half of 24

5. Is 6 : 10 equivalent to 9 : 15?

6. Ava changes 3 : 7 to 5 : 9. Is it the same ratio?

7. Find $3 \times \frac{2}{3}$.

8. Find $\frac{1}{2} + \frac{1}{3}$.

9. $4.5 + 0.25$

10. A box is 4 cm by 3 cm by 3 cm. Find its volume.

11. 7×2

12. A box is 7 cm by 2 cm by 3 cm. Find its volume.

13. A tile bag has red : blue = 3 : 2. The total is greater than 20 and less than 30. How many of each color?

14. Red: blue is 6 : 5. Complete (6, 5), (36,__).

15. A mix has 10 raisins and 20 nuts. Divide both counts by 5; write the new ratio.

16. A recipe uses 4 cups oats for 8 cups seeds. How many cups seeds go with 6 cups oats?

17. Gold: silver beads are 3 : 5. There are 40 beads. Find each count.

18. Orange: purple tiles are 5 : 3. What fraction of all tiles is orange?

19. Complete the ratio table: (5, 7), (25,__), (__, 42).

20. Is 4 : 6 equivalent to 2 : 4? Explain without using the difference.

21. Mix A has 3 cups fruit for 5 cups yogurt. Mix B has 7 for 10. Which has more fruit for the same yogurt?

22. For each box there are 4 pencils. Write ordered pairs for 0, 4, and 5 boxes. Is (0, 0) included?

23. 8 bags each hold $\frac{3}{6}$ kilogram. Find the total mass.

CHALLENGE / Explain your plan and check the result

C1. A box has 72 beads with red: blue 4 : 5. Without adding or removing beads, recolor some blue beads red to make equal counts. How many must be recolored?

C2. The pair (n, 12) has ratio 2 : 3. Find n. Then test whether adding 6 to both entries preserves the ratio. Give a correct equivalent pair with first entry 14.